

Exercise 13&14: (Pumping Lemma for NonCFL)

Prove by P.L. that the following languages are not CFL.

$$1. L1 = \{a^n b^{n+1} c^{n+2} : n \geq 0\}$$

Proof: Assume that L1 is CFL. There is PDA accepts L1 with m number of states

1) We pick a string $w = \underline{a^m b^{m+1} c^{m+2}}$, $|w| \geq m$

2) Opponent decomposes w into $uvxyz$, $|vxy| \leq m$, $|vy| \geq 1$.

$$2.1) \quad \begin{array}{c} \overbrace{aa \dots a}^m \quad \overbrace{bb \dots b}^{m+1} \quad \overbrace{cc \dots c}^{m+2} \\ | \text{---} u \text{---} vxy \text{---} | \text{-----} z \text{-----} | \end{array} \quad \text{Let } vy = a^k; \quad k \geq 1$$

We pick $i=2$, $a^{m+k} b^{m+1} c^{m+2} \notin L1$ since $k \geq 1$.



2.2) $\overbrace{aa \dots a}^m \overbrace{bb \dots b}^{m+1} \overbrace{cc \dots c}^{m+2}$
 $| \text{---} u \text{---} | \text{---} vxy \text{---} | \text{---} z \text{---} |$

2.3) $\overbrace{aa \dots a}^m \overbrace{bb \dots b}^{m+1} \overbrace{cc \dots c}^{m+2}$
 $| \text{---} u \text{---} | \text{---} vxy \text{---} | \text{---} z \text{---} |$

2.4) $\overbrace{aa \dots a}^m \overbrace{bb \dots b}^{m+1} \overbrace{cc \dots c}^{m+2}$
 $| \text{---} u \text{---} | \text{---} v \text{---} | x | \text{---} y \text{---} | \text{---} z \text{---} |$

Similar analysis to case 2.1.

Let $|v| = k_1$, $|y| = k_2$, $k_1 + k_2 \geq 1$

→ We pick $i=0$, $a^{m-k_1} b^{m+1-k_2} c^{m+2} \notin L_1$ since $k_1 + k_2 \geq 1$.

2.5) $\overbrace{aa \dots a}^m \overbrace{bb \dots b}^{m+1} \overbrace{cc \dots c}^{m+2}$
 $| \text{---} u \text{---} | \text{---} v \text{---} | xy \text{---} | \text{---} z \text{---} |$

We pick $i=2$, $a^m b^{k_2} a^{k_1} b^{k_2+m+1} c^{m+2} \notin L_1$.

2.6) $\overbrace{aa \dots a}^m \overbrace{bb \dots b}^{m+1} \overbrace{cc \dots c}^{m+2}$
 $| \text{---} uvx \text{---} | \text{---} y \text{---} | \text{---} z \text{---} |$

Similar analysis to case 2.5.

2.7) – 2.9) vxy overlaps b^{m+1} and c^{m+2} which have similar analysis to 2.4) – 2.6).

3) We can reach contradiction with P.L. in all cases. Therefore, L_1 is not CFL.

$$2. L2 = \{ w \in \{a, b, c\}^* : n_a(w) = \min(n_b(w), n_c(w)) \}$$

Proof: Short version (You suppose to show a full version in the exam).

$$w = a^m b^m c^m$$

m

aa ... a

m

bb ... b

m

cc ... c

|---vxy---

$$i = 0, 2, 3, \dots$$

Case 2)

|---vxy---

$$i = 0$$

Case 3)

|--vxy--|

$$i = 0$$

If $k_1 \neq 0$ AND $k_2 \neq 0$, use $i=2, 3, \dots$

If $k_1 = 0$ AND $k_2 \neq 0$, use $i=0$

If $k_1 \neq 0$ AND $k_2 = 0$, use $i=0, 2, 3, \dots$

Case 4)

vxy overlaps $a^m b^m$, $v = a^{k_1}$, $y = b^{k_2}$

Case 5)

vxy overlaps $a^m b^m$, v overlaps $a^m b^m$, $y = b^k$

$i = 2, 3, \dots$

aaabab is not allowed

Case 6)

vxy overlaps $a^m b^m$, $v = a^k$, y overlaps $a^m b^m$

$i = 2, 3, \dots$

Similar to 5

Case 7)

vxy overlaps $b^m c^m$, $v = b^{k_1}$, $y = c^{k_2}$

Similar analysis to Case 4.

Case 8)

vxy overlaps $b^m c^m$, v overlaps $b^m c^m$, $y = c^k$

$i = 0, 2, 3, \dots$

bcbccc is not allowed

Case 9)

vxy overlaps $b^m c^m$, $v = b^k$, y overlaps $b^m c^m$

$i = 0, 2, 3, \dots$

Similar to 8